

Pharmacokinetics (PK) ADME Processes

Practice Quiz

Some items may have more than one answer.

Goals of Practice Quizzes:

1. To help students identify what they know and what they understand by applying pharmacology knowledge to clinical case vignettes.
2. To help students get used to thinking through the “Board-style” case vignettes and questions. NYITCOM exams, COMLEX and USMLE test students’ medical knowledge using these types of questions.



CHALLENGE: Cover the options and try to answer the questions without looking at the choices.

Spaced retrieval, mixing up topics, elaborating, and drawing figures, diagrams, and flow charts are learning techniques for successful learning

My recommendation: Home in on the information you need to answer the question, which, at this time, is related to drug pharmacodynamics. *Don’t let yourself get distracted by the background information unless it is relevant for answering the question.*

HOW CAN I FIND THE ANSWERS? “Search” my Notes Handout. (You do not need to read the whole document if you do not wish to.)

All answers can be found by doing a “search” of my Notes Handout *or* the textbook chapters designated in the resources of each question (links provided).

➤ **Pharmacology bridges physiology, pathophysiology, and clinical practice.**

THE ANSWER KEY WILL BE POSTED THE FRIDAY NOVEMBER 21.

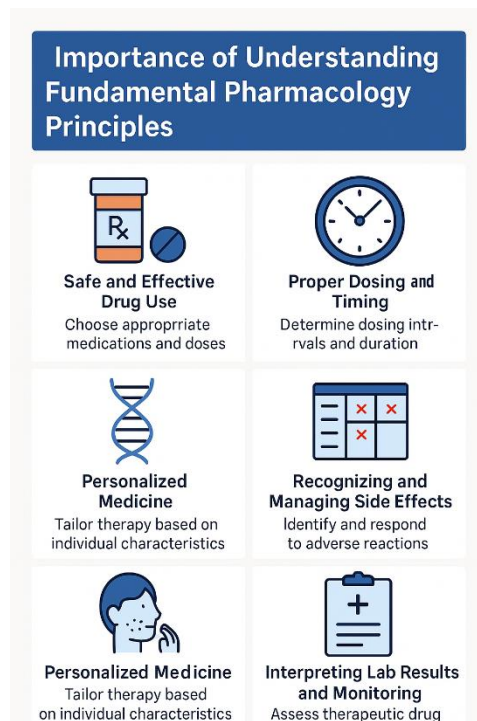
Questions are below.

1. Why is it important for clinicians to understand fundamental pharmacology principles?
Your own reflection.

Rationale: Optimization of therapy for the individual patient

Tailoring therapy to the genetics, lifestyle preferences, and comorbidities of the individual patient aims to provide preventive and therapeutic strategies that are more effective, safer, and ultimately result in more successful health outcomes.

The following image was generated by AI (Copilot) and reviewed by me.



2. A pharmaceutical company developed a drug for the treatment of glioblastoma (brain tumor) that can be taken orally. It is a prodrug well absorbed from the gastrointestinal (GI) tract by passive diffusion. It is nonezymatically converted to the active anticancer metabolite, MTIC, which produces high systemic blood levels (100% bioavailability). It readily penetrates the blood-brain barrier and distributes to the site of action. MTIC is not a substrate of any plasma membrane proteins. The drug is excreted mainly in the urine as inactive compounds.

Which one of the following options BEST describes the physicochemical characteristic that permits the movement of the active metabolite within the body to reach the target of action?

- A. The drug has a chiral center.
- B. The drug has a high lipid:water coefficient.*
- C. The drug forms electrostatic bonds with host structures.

- D. The drug is a small molecule chemical.
- E. The drug is a substrate of solute carrier (SLC) transporters.

Hint: Page 3 of Notes Handout.

Rationale: Lipid:water partition coefficient is the ONE BEST ANSWER.

Thinking it through:

Point 1, the parent drug: The vignette indicates that the parent molecule is an oral prodrug (no activity of its own) that passively diffuses across the lipid barriers of GI epithelium.

This statement infers that the drug is lipid soluble – it has a high lipid:water partition coefficient – which enables it to passively diffuse across lipid bilayers.

Point 2, the active metabolite: To reach the tumors in the brain, the active compound – the active metabolite MTIC in this example – must cross the blood-brain barrier, which is formed by tight junctions between endothelial cells of the capillary cell walls, pericytes in the capillary basement membrane, and astrocyte foot processes.

It is stated that the anticancer compound MTIC “readily penetrates the blood-brain barrier” and is not a substrate of membrane proteins. The student can deduce that the drug is not crossing the BBB via membrane transporters (or being effluxed by P-glycoprotein). Therefore, the drug molecule MUST BE lipid soluble.

PHARMACOLOGY PRINCIPLES Learning points

Physicochemical properties: Physicochemical characteristics determine the movement of a drug across physiologic barriers.

A drug molecule must have the appropriate size, unique shape, atomic composition, electrical charge, chirality (stereoisomerism), degree of ionization, and relative lipid solubility and stability to be able to:

- cross physiologic barriers (membrane permeability)
- reach the site of action, and
- bind to its target.
- A practical drug should be inactivated and excreted at a reasonable rate to:
 - control dosing
 - control the duration of action of the drug
 - prevent toxicity due to accumulation of drug
 - maintain efficient clearance (homeostasis) by preventing overload of detoxification systems (mainly liver and kidneys)

Chemistry review: Lipophilicity of a drug is characterized by the lipid:water partition coefficient – the ratio of the concentration of drug in two immiscible solvents at equilibrium: lipid and water.

It compares the solubilities of the solute (drug in pharmacology) in the two liquids (frequently octanol, a fatty alcohol, and water).

Passive diffusion and drug distribution: A drug passively moves across membranes down its concentration gradient.

- Most drugs distribute through the body by passive diffusion, which is efficient, energy independent, and not saturable.
- ***Passive diffusion between compartments favors small, lipophilic, uncharged molecules.***
For context (no need to memorize): The ideal size appears to be between 100-1000 daltons (unified atomic mass units).
 - Drugs with high molecular weights (>1000, usually proteins) do not freely diffuse between body compartments.
 - Many monoclonal antibody (mAb) drugs (~145,000 daltons) and other proteins have been marketed since 1994.
 - mAbs act on extracellular targets – circulating proteins or plasma membrane receptors.

Biology review: Passive diffusion across membranes is directly proportional to the:

- magnitude of the concentration gradient across the membrane,
- the lipid:water partition coefficient of the drug, and
- the membrane surface area exposed to the drug.

Drug Distribution and dynamic equilibrium:

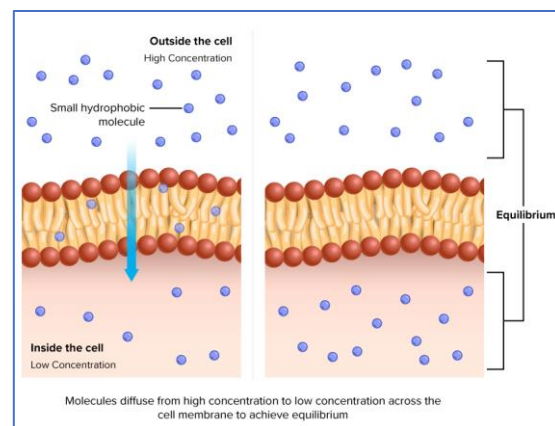
- Drugs move from the plasma circulation into tissues until a dynamic equilibrium is reached (with repeated doses).
- When drug administration is withdrawn and the drug is eliminated (metabolism and or excretion), plasma levels decrease. Then, drug moves from tissues (higher concentration) into the blood (lower concentration) until the drug is eliminated from the body by metabolism and/or excretion.

❖ **Definition of Dynamic Equilibrium (drugs):**

A state in which drug molecules in the system move back and forth between blood and tissue (or between the intracellular space and the extracellular fluid), and the rate of movement in each direction is equal, resulting in *no net change in concentration* on either side.

Dynamic equilibrium (steady state concentration) is typically reached over time when a drug is administered repeatedly at intervals equal to its half-life.

Steady state and half-life are explained in the Clinical Pharmacology lecture.



<https://www.ck12.org/flexi/life-science/diffusion/what-happens-at-equilibrium-in-diffusion/>

Incorrect options:

- A. A chiral molecule exists in two stereoisomers that are mirror images of each other. *The enantiomers of a drug usually have different potencies and effects.*
- C. Drugs interact with receptors by electrostatic, hydrophobic, and covalent bonds.
Interactions by electrostatic bonds are common.
Hydrophobic bonds are weak and probably important in the interactions of highly lipid-soluble drugs with cell membranes and the internal receptor “pockets”.
Covalent bonds are strong and irreversible.
- D. Small molecule chemical drugs with favorable physicochemical properties can be orally absorbed. Biologic drugs are large peptides and proteins, usually antibodies and fusion proteins, are not absorbed from the GI tract because they are broken down by intestinal proteases.
- E. Membrane impermeant drugs (drugs that do not passively diffuse across lipid-bilayers) may flow down their concentration or electrochemical gradient via facilitative transporters, a subfamily of SLC transporters.

Reference: PK parameters: Lexidrugs monograph: Temozolomide

3. Methamphetamine is weak base with a pKa of ~10. Approximately 33% of the dose is excreted in the urine as unchanged (unmetabolized) drug.

i) Which of the following would REDUCE the concentration of the drug in the excreted urine?

A. Urine pH 5.5

B. Urine pH 7.0

ii) Name and write out the well-known equation that explains your answer?

Hint: Pages 4-5 on the Notes Handout

Rationale: B: Urine pH 7.0 / Henderson-Hasselbalch equation

Higher $[B]_{\text{urine}}$ will be present at higher pH_{urine} , which will reduce the concentration of drug excreted in urine.

Henderson-Hasselbalch equation for determining the ratios of unprotonated/protonated forms of weak acid and weak base:

For weak acid: $pH = pK_a + \log ([A^-]/[HA])$

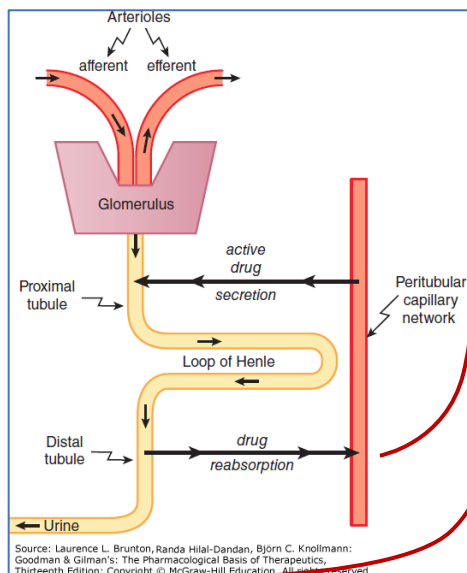
For weak base: $pH = pK_a + \log ([B]/[BH^+])$

General Rule

for a weak acid:	for a weak base:
If $pH = pK_a$, $[A^-] = [HA]$	If $pH = pK_a$, $[B] = [BH^+]$
If $pH < pK_a$, $[A^-] < [HA]$	If $pH < pK_a$, $[B] < [BH^+]$
If $pH > pK_a$, $[A^-] > [HA]$	If $pH > pK_a$, $[B] > [BH^+]$

Methamphetamine example: Ion trapping of weak bases in urine:

Weak bases in the unprotonated B form are more lipophilic and able to passively diffuse across the renal epithelial cells back into the blood stream.



Reabsorption: The higher the urine pH, the greater the concentration of a weak base in the B form ($[B]$), the greater the passive reabsorption of the drug from the nephron back into the blood.

Ion trapping: At steady state (dynamic equilibrium), as a weak base diffuses from a medium with a higher pH, such as blood (pH 7.4), into a more acidic medium, such as urine, which is usually acidic, a greater fraction of the base will be in the ionized BH^+ form. BH^+ cannot back diffuse into the blood – it remains “trapped” in the urine and will be excreted.

It is the opposite for weak acids. A greater concentration of the ionized A^- form will be trapped in more alkaline urine.

Learning point: Determinants of passive diffusion of weak acids and bases across physiologic barriers.

- pKa: Only the non-ionized species is lipophilic (will passively diffuse across lipid barriers).
- The pKa of the drug and the pH of the environment influence the state of ionization of the molecule (ionized or non-ionized) on either side of the membrane.
- The pKa and pH determine the concentration of ionized drug that will be trapped on one side of the membrane.

4. A 55-year-old woman takes a diuretic for management of hypertension. The drug blocks sodium reabsorption through a specific ion channel in renal epithelial cells. Sodium and water are excreted (natriuresis and diuresis), lowering blood volume, which lowers blood pressure.

The physiological function of the specific ion channel is to allow movement of sodium down its electrochemical gradient into the epithelial cell, which is then reabsorbed into the bloodstream. Chloride (Cl^-) is drawn along using the energy from the sodium ion electrochemical gradient. Which one of the following best describes the mechanism of the Cl^- movement?

- A. Passive diffusion
- B. Aqueous diffusion
- C. Paracellular transport
- D. Facilitated diffusion

E. Secondary active transport*

F. Primary active transport

Hint: Notes p7-9

Rationale: E. secondary active transport via a symporter.

The vignette describes the renal Na^+, Cl^- cotransporter (NCC), which is a symport in the solute carrier (SLC) superfamily of transporters.

Learning point: Modes of drug movement across physiologic barriers.

Be able to describe the mechanisms of SLC and ABC transporters.

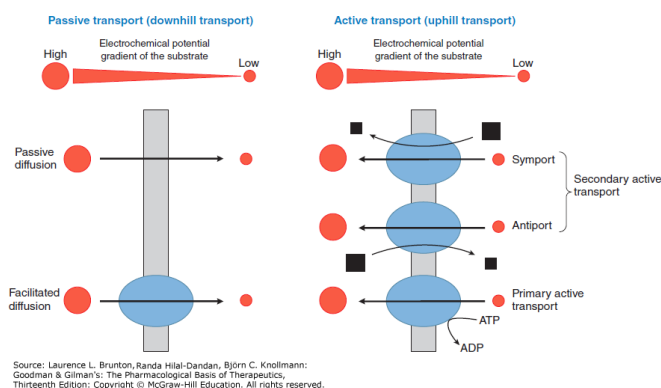
SLC superfamily of transporters move ions and polar molecules across cell membranes by various means:

Facilitated transport: uniport

- A transporter that moves a single substance (uniport) down its concentration gradient (passive diffusion through the channel)
 - Several glucose transporters and amino acid transporters are examples carrier proteins that are uniports.

Secondary active transport: symports and antiports

- Transporters that move two or more substrates across the membrane.
- Movement of one substrate (such as sodium) down its electrochemical gradient provides the energy for the simultaneous movement of the other substance against its concentration gradient.
- Secondary active transport *does not use ATP* to power the transport but does require a sodium pump (Na^+/K^+ ATPase) to establish the sodium gradient.



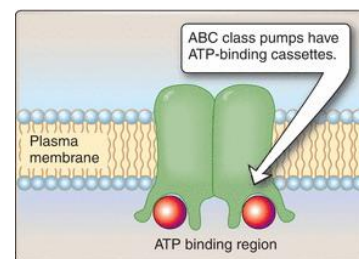
ABC superfamily: Primary active transport

ABC transporters use energy released by ATP hydrolysis to pump substances, including drug substrates, *against* a concentration gradient.

P-glycoprotein is a prime example of an ABC transporter.

What is the mechanism?

The ATPase in each binding cassette on the intracellular side of the ABC transporter binds an ATP molecule, bringing it near a water molecule, which facilitates a nucleophilic attack of the water molecule on the phosphate group of ATP, breaking the bond, releasing ADP + Pi + energy.



The energy induces a conformational change in the transmembrane domains. The transporter switches from the inward-facing to the outward-facing conformations, facilitating substrate translocation.

The ATPase returns to its original conformation, ready to bind another ATP molecule.

(FYI: There are various types of ATPases in the body and diverse functions.)

- **Practice point: *Transport proteins are saturable and can be inhibited or induced by drugs***, which can increase or decrease the plasma concentrations of affected drugs. Understanding this mechanism of drug interactions is critical for optimizing therapy for the individual patient.

Incorrect options:

- A. Passive diffusion is the movement of a substance from an area of high concentration to an area of low concentration without the use of energy or assistance from carrier (transport) proteins.
Lipid soluble drugs cross lipid bilayers driven by their concentration gradients. This type of drug movement is *not energy dependent and not saturable*.
- B. Aqueous diffusion is the movement of substances dissolved in water from an area of higher concentration to an area of lower concentration within the water-based solution.
- C. Paracellular transport is the movement of solutes between cells down their concentration or electrochemical gradient. Specific transporters are not involved.
- D. Facilitated diffusion is the passive movement of substances across physiologic barriers down their concentration gradient via carrier proteins of the SLC superfamily.
- F. Primary active transporters use energy from ATP hydrolysis to power the movement of ions or molecules against a concentration gradient.

Additional minor mode of drug transport: In endocytosis, a molecule is engulfed by the cell membrane and released inside the cell in an endosome.

5. Referring to the vignette above, it is stated that chloride (Cl^-) is drawn along using the energy from the sodium ion electrochemical gradient.

i) By what mechanism is the electrochemical gradient generated?

ii) What category of membrane transport generates the electrochemical gradient?

Rationale:

i) Primary active transport using ATPase to power the transport of 3 Na⁺ out of the cell and 2K⁺ into the cell

ii) ATP binding cassette (ABC) superfamily of transporters

The three main categories of membrane transport are:

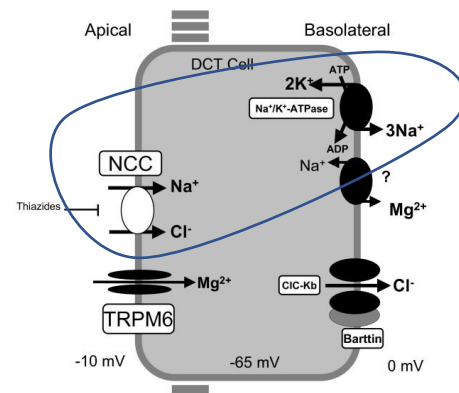
Primary active transport (uses energy, ATP)

Facilitated diffusion

Secondary active transport

Please see the explanation in the Rationale for Question #4.

The Na⁺/K⁺ ATPase is a transmembrane ion transporter present in all animal cells and has many functions.



Attribution:

https://www.researchgate.net/publication/23137480_Gitelman_syndrome?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Il9kaXJlY3QiLCJwYWdlIjoieX2RpcmVjdCJ9fQ

6. A 52-year-old man diagnosed with atrial fibrillation is provided an e-prescription for an anticoagulant called apixaban to reduce formation of blood clots due to the stasis of blood in the left atrium. The drug is a substrate of the proteins encoded by the *ABCB1* gene and *CYP3A4* gene.

If the patient were to require antimicrobial therapy with a drug that inhibits the proteins expressed by these genes...

i) ...what would be the major potential adverse event related to the anticoagulant therapy?

ii) ...what is the mechanism of the drug interaction that can cause the adverse effect?

Hint: Notes p7 and p23

Rationale:

i) Bleeding

ii) The *CYP3A4* gene encodes CYP3A4 metabolic enzyme.

The *ABCB1* gene encodes P-glycoprotein efflux transporter.

- ❖ The two proteins are frequently co-expressed in tissues like the small intestine and the liver and have overlapping substrate specificity. They work together in eliminating xenobiotics like drugs and toxins.

Anticoagulants interfere with clot formation – a concentration-response relationship

Anticoagulants are commonly prescribed to prevent formation of thrombi that could result in a cardiovascular adverse events: stroke, heart attack, deep vein thrombosis, pulmonary embolism.

- Recall that the concentration of drug in plasma determines the concentration of drug at the site of action, which determines the magnitude of effect.
- Higher plasma levels of the anticoagulant increase risk of bleeding – the greater the concentration, the higher the risk.

If the bleeding occurs in the brain, the patient could suffer a hemorrhagic stroke.

Mechanism of the drug-drug interaction:

Drug A = the substrate, apixaban in the case vignette

Drug B = the drug added to the patient's medication regimen, antimicrobial drug

When Drug A is taken alone:

- a fraction of the dose of Drug A would be pumped back into the intestine by P-glycoprotein and a fraction of the dose would be metabolized to inactive metabolites by CYP3A4.
- The fraction of the dose that survives these processes and reaches the systemic circulation is the bioavailability of the drug. When taken regularly as directed, Drug A will achieve stable therapeutic blood levels.
- Drug developers determine the doses that will achieve safe therapeutic effects by conducting clinical trials.

When Drug B is added to the patient's medication regimen, it inhibits P-glycoprotein and CYP3A4:

- Inhibition of P-glycoprotein by Drug B would prevent efflux of Drug A back into the lumen / into the biliary system / into the kidney → increases plasma levels of the drug
- Inhibition of CYP3A4 metabolism reduces the rate of metabolism of Drug A → a greater concentration of Drug A would reach the systemic circulation → higher plasma levels of the anticoagulant

Result: Therefore, by both mechanisms, a greater amount of Drug A could reach the systemic circulation.

MUST KNOW

P-gp's mechanism of action, physiologic function, mechanisms of drug interactions, and function in tumor and microbial resistance to drug therapy.

Inhibition of P-gp and overexpression in microbial cells and tumor cells **significantly impacts drug therapy.**

NOTE: Many drugs are P-glycoprotein and CYP3A4 inhibitors.

But...not all drugs are substrates of P-glycoprotein.

Drugs that cross physiologic barriers by passive diffusion and are not pumped out, or pass into the cell by SLC transporters, can reach the internal sites listed above.

TAKEAWAY: P-glycoprotein is expressed in many tissues:

Tissue expression	Mechanism for reducing drug concentrations Drug substrates of P-gp are:
Intestinal epithelial cells	Pumped out of intestinal epithelial cells back into the intestinal lumen, reducing absorption.
Liver hepatocytes	Excreted into biliary system, then back to the intestinal lumen
Kidney	Facilitates secretion of drugs into the nephron for excretion
Blood-brain barrier	Prevents drug from passing through capillary endothelial cells, limiting entry into the central nervous system by
Blood-testis barrier	Protects developing sperm cells from harmful substances
Blood-placenta barrier	Prevents drugs from entering the fetal circulation
Blood-retina barrier	Helps maintain the tightly regulated environment of the retina
P-glycoprotein is expressed in many other tissues both on the plasma membrane and on various intracellular membranes.	
Mechanisms of resistance related to P-glycoprotein and other efflux pumps	
Bacteria and fungi	Drug efflux from microbial cells leads to resistance to antimicrobial drugs. This means the organism will continue to flourish despite the presence of the drug. Microbial resistance significantly complicates therapy.
Tumor cell	P-gp is overexpressed by many types of tumor cells, resulting in resistance to the anticancer drug(s). Reduced drug effectiveness can lead to treatment failure and tumor recurrence. P-glycoprotein is also called multidrug resistance protein 1, MDR1.

Incorrect options:

- A. ATP phosphatase: A phosphatase is a type of hydrolase that removes phosphate from molecules, including ATP. It is not a transporter.
- B. Channel protein: These pore-forming membrane proteins allow specific ions to move passively down electrochemical or concentration gradients.
- C. Cytochrome P450 isozyme: CYPs are Phase 1 oxidases (mono-oxygenases) involved in metabolizing a wide-range of lipophilic substrates, including a wide-range of drugs.
- D. G protein: Guanine nucleotide proteins (G proteins) are signal transducers. They act as molecular switches that activate or inhibit enzymes (such as adenylyl cyclase and guanylyl cyclase), which, in turn, trigger second messengers and cellular responses.
- E. Transferase: Phase 2 metabolic enzymes transfer a molecule from a cofactor and attach it to the substrate, including drugs and drug metabolites, increasing the hydrophilicity of the substrate.

7. A 74-year-old man with a history of rheumatoid arthritis and a recent diagnosis of advanced pancreatic cancer presents to the ED with altered mental status and symptoms and signs of acute salicylate toxicity. His wife found an empty bottle of regular strength aspirin near him and believes he might have tried to kill himself. She notes that he has seemed depressed for about the last month since receiving his cancer diagnosis.

The patient is given respiratory and cardiovascular support, gastric lavage and activated charcoal, and glucose supplementation. Intravenous sodium bicarbonate (NaHCO_3) is titrated to a urine pH of 7.5 to 8. Serum salicylate and serum potassium levels are monitored.

What is a goal of sodium bicarbonate administration in this patient? Short phrase.

Hints: Aspirin's chemical name is acetylsalicylic acid. It has a pK_a of ~ 3.5 .

Urine pH is usually slightly acidic, 6.0. (typical range: 5.5-7.5; normal range 4.5-8.0)

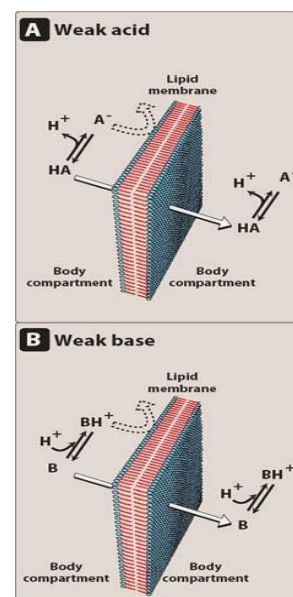
Rationale: Alkalization of the urine

Increasing urine pH increases salicylate concentration in urine (ion trapping) → increases clearance of the salicylate.

Here's how:

Salicylates are weak acids. In an acidic urine, a greater fraction of the drug is in the non-ionized HA form, which passively diffuses through renal cell membranes (is reabsorbed) into the blood stream and tissues.

Ion trapping effect: Increasing the pH of the urine to >7.5 increases the fraction of the drug in the ionized A^- form: “traps the ion” in the urine. There is, therefore, a greater concentration of drug excreted in alkaline urine than in acidic urine, facilitating clearance of the salicylate.



Summary: Alkalinization increases salicylate $[\text{A}^-]_{\text{urine}}$ concentrations (concurrently reducing $[\text{HA}]_{\text{urine}}$ concentrations) → increases salicylate clearance.

Bicarbonate also increases the systemic pH to counteract metabolic acidosis.

Definitions:

Weak acids and weak bases do not completely dissociate into their constituent ions in aqueous solution.

This means that a weak acid does not completely dissociate into its hydrogen ions H^+ (protons), and its conjugate base, A^- . A weak base does not completely dissociate into its hydroxyl ions, OH^- , and conjugate acid, BH^+ .

Learning point:

Understanding pKa helps in predicting absorption and distribution of a drug and in managing toxicity.

Many drugs are weak acids or weak bases

1. Weak acids and weak bases are present in solution as both the nonionized, lipid-soluble, diffusible form and the ionized (charged), polar, poorly diffusible form.
2. Physiologic pH is ~7.4. The degree of ionization based on the pKa of the weak acid or weak base determines its concentration in plasma, interstitial fluid, intracellular space, and at the site of action.
3. This case vignette is intended to help students apply the concept of ion trapping in the clinical context of increasing urine excretion to increase clearance of a toxic concentration of drug.
4. Remember, urine pH also affects clearance of therapeutic doses of weak acids and weak bases.

8. A PGY1 medical resident on his way to work in the free clinic finds an unresponsive young female in a doorway. He detects pinpoint pupils, respiratory rate of 8 breaths per minute, and needle tracks on her arms and legs (findings are consistent with opioid overdose). He quickly administers the antidote (opioid receptor antagonist), which he always carries with him. The antidote is rapidly absorbed through mucous membranes and bypasses first-pass effect. It has an onset of action in as little as 3 minutes, a peak concentration in about 20-30 minutes, and a duration of action of up to 2 hours.

i) By what route of administration was the opioid antagonist most likely administered to the unconscious patient?

- A. Intramuscular
- B. Intranasal*
- C. Oral
- D. Rectal
- E. Subcutaneous
- F. Transdermal

Hint: The pharmacokinetic properties will give you the clue to the correct option.

ii) What is the drug, if you happen to know?

Rationale: B. intranasal

Thinking it through:

- The vignette states that the drug is absorbed through mucous membranes AND bypasses first-pass effect (first-pass metabolism) AND has a relatively rapid onset of action.

- These possibilities are
 - 1) absorption across nasal mucous membranes (intranasal administration),
 - 2) through the mucous membranes of the cheek (buccal administration), and
 - 3) under the tongue (sublingual).
 Drugs administered by these routes are absorbed directly into the systemic (central) circulation and bypass first-pass metabolism / first-pass effect.
 Intranasal is the only one of the three given as an option.

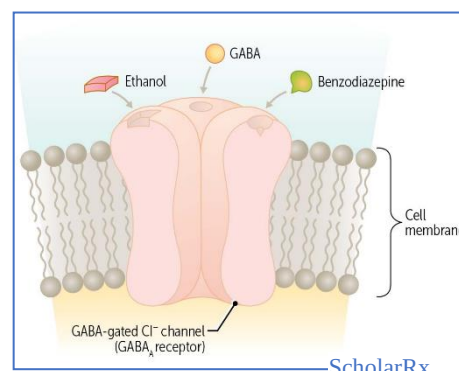
Incorrect options:

- Intramuscular injection administers drug directly into the muscle.
Drug molecules in aqueous solution are rapidly absorbed through the capillaries directly into the blood stream.
- Oral drugs are absorbed across the mucosal barrier of the GI tract into the portal circulation and are transported directly to the liver, where they are subject to first-pass metabolism. Onset of oral drugs is delayed (lag time) because the drug must pass through the stomach (not an absorptive organ), be released from the dosage form, and be absorbed in the small intestine.
- Rectal administration (lower rectum) bypasses first-pass metabolism, somewhat, because about 50% of the drainage of that rectal region bypasses the portal circulation.
- Subcutaneous injection administers drug into the tissue layer between the skin and the muscle.
- Transdermal administration bypasses the first-pass effect by moving into the systemic circulation across the skin barriers, not mucosal barriers. Onset of drug action is slow because of slow-release formulations and passage of drug through the dermal layers.

9. A 20-year-old male being treated for methamphetamine toxicity is given a benzodiazepine for sedation. The sedative passively diffuses through cell membranes, moves across the blood-brain barrier, and binds to GABA-A receptors. Which one of the following BEST describes the physicochemical characteristic that permits this drug's movement across physiologic barriers to reach its target of action?

- A. Lipophilicity*
- B. Stereoisomerism
- C. Structural similarity to fatty acid chains of lipid bilayer
- D. Tendency to form hydrophobic bonds

Hint: Notes Part 2, page 4



Rationale: A. Lipophilicity

The one BEST answer. The other options are structural characteristics that may contribute to the lipophilicity of the drug.

Physiologic barriers consist of membrane phospholipid bilayers. To cross through membranes, drugs must be able to dissolve into the lipid rich membrane – they must have lipophilicity – and be nonpolar (or minimally polar) and uncharged at physiologic pH. Smaller molecules diffuse more easily than larger ones.

Incorrect options:

- Stereoisomerism describes the shape of molecules. Stereoisomers are molecules having the same molecular formula and sequence of bonded atoms but differ in the spatial arrangement of their atoms.
- Structural similarity to fatty acid chains of lipid bilayer is not necessary for drugs to pass through membranes. The lipid:water partition coefficient is the indicator of a molecule's ability to pass through.
- Tendency to form hydrophobic bonds is important for a drug's ability to bind to its target. It may also be involved in the passage through the lipid bilayers.

10. A 4-year-old child is admitted for the treatment of bacterial meningitis with antibiotics.

Which one of the following is the most appropriate route of administration in this case?

- A. Intra-arterial
- B. Intramuscular
- C. Intraosseous
- D. Intrathecal
- E. Intravenous*
- F. Intraventricular
- G. Oral

Definition: Meningitis is an inflammatory disease of the tissues surrounding the brain and spinal cord. Bacterial meningitis reflects an infection of the arachnoid mater and cerebral spinal fluid (CSF) in the subarachnoid spaces and ventricles.

Hints 1: In bacterial meningitis, the blood-brain barrier and the blood-CSF barrier become leaky due to the inflammation.

Hint 2: Bacterial meningitis is life-threatening.

Hint 3: Notes handout pp.9-13 | PowerPoint slides 53-57.

Intent of this question: To help you start thinking critically and clinically when presented with patient information.

Rationale: E. intravenous

Even without having a lecture on it, I'm sure you recognize that inflammation of the tissues surrounding spinal and brain tissues is a potentially life-threatening situation.

Definition, meninges: dura mater, arachnoid mater, pia mater

- 1) Life-threatening diseases must be treated quickly and with adequate, often high, doses → high systemic concentrations of drug. Treatment is multifaceted and often multidisciplinary.
- 2) Regarding drug therapy for meningitis, ideally a drug that rapidly penetrates the CSF in therapeutic concentrations is desirable.
- 3) Importantly, a drug that is highly effective against the potential causative microbes AND is not toxic to brain tissues are essential criteria in drug selection.
- 4) The trick is to match the three criteria – therapeutic drug levels in CSF, efficacy, and safety.

The questions become:

What drugs are effective?

Can they penetrate into the CSF when given intravenously?

The IV route is ideal because of 100% bioavailability, high plasma levels can be achieved, rapid onset of action, predictable plasma levels, and relatively easy intravascular access.

What are the safety profiles of the effective drugs?

The hint provided with the clinical vignette indicates that the blood-brain barrier and the blood-CSF barrier become leaky due to the inflammation. Therefore, substances that would normally be prevented from entering the CNS are now able to pass through from the systemic circulation into the CNS.

Thus, a drug that achieves high systemic blood levels would likely achieve therapeutic CSF levels.

Always check the evidence. Never assume.

Thus, intravenous administration is the preferred route in this case.

Intraosseous is an alternative if vascular access cannot be obtained.

- **Practice point:** Once a patient is stabilized and out of danger, a switch from parenteral administration to oral administration is often appropriate.

Incorrect options:

- **Intra-arterial** administration is for localized effect of the drug in a particular organ or tissue, *ie* liver tumors, head and neck cancers, or diagnostic agents.
Only skilled personnel should perform intra-arterial injection. Inadvertent arterial administration can cause serious complications.
- **Intramuscular** injection provides generally rapid (not as rapid as IV) and uniform absorption. CNS infections, like meningitis, require higher doses of antibiotic to achieve

therapeutic concentration in the cerebral spinal fluid. In this type of case, IM injection is not preferred.

- **Intraosseous** infusion directly into the highly vascularized bone marrow, which drains into the venous system, is a rapid technique for achieving therapeutic plasma concentrations. This route of administration is useful for acute, life-threatening emergencies, especially in children, when standard venous access methods have failed, or as the first attempt at vascular access in full cardiopulmonary arrest or severe shock.
 - **Intrathecal** administration bypasses the blood-brain barrier by injecting drugs directly into the spinal subarachnoid space.
Disadvantages are potential for infection and administration by trained personnel.
 - **Intraventricular** administration bypasses the blood-brain barrier by injecting drugs directly into ventricle in the brain for the treatment of brain tumors or severe CNS infection.
Disadvantages are potential for infection and expertise in intraventricular technique.
 - **Oral** absorption from the GI tract is relatively slow. Peak concentrations are achieved typically in 1-2 hours. Also, high enough plasma concentrations may not be achievable with oral administration.
-

11. A 52-year-old male being prepared for hospital discharge following treatment of deep vein thrombosis is evaluated for outpatient anticoagulant therapy. Medical history is significant for seizure disorder treated with carbamazepine (a strong CYP3A4 inducer). An oral factor Xa (pronounced ten-a) inhibitor would be effective and convenient for the patient but is contraindicated in this case.

i) What is an adverse drug event that could occur with the concomitant use of the carbamazepine and the oral factor Xa inhibitor?

- A. Hemorrhagic stroke (bleeding in the brain)
- B. Pulmonary embolism (thrombus breaks off and travels to the lungs)*

ii) What is the mechanism of this drug-drug interaction?

Short phrase

Rationale:

i) Pulmonary embolism

ii) Increased rate of metabolism of the oral factor Xa inhibitor that reduces plasma levels (bioavailability) of the anticoagulant decreases plasma levels of the drug – to possibly below the therapeutic range, which reduces anticoagulant effectiveness.

Strong CYP3A4 inducers (rifampin, carbamazepine, phenytoin) significantly reduce plasma levels of CYP3A4 substrates. These drug combinations should be avoided.

General Mechanism —the takeaway—	Oral Factor Xa example —the application—
induction of CYP genes ↓ ↑ mRNA expression ↓ ↑ CYP protein synthesis ↓ ↑ rate and extent of drug metabolism ↓ ↓ drug bioavailability ↓ ↓ therapeutic effect ↓ ↑ risk of an adverse event related to the condition the drug is intended to treat	Induction of CYP3A4 ↓ ↑ expression of mRNA ↓ ↑ CYP3A4 synthesis ↓ ↑ rate and extent of anticoagulant metabolism by CYP3A4 ↓ ↓ anticoagulant levels in plasma ↓ ↓ coagulation inhibition ↓ ↑ risk of pulmonary embolism (life-threatening)

The mechanism of the drug-drug interaction

- Certain drugs like carbamazepine bind nuclear receptors in liver cells, which activates the transcription of cytochrome P450 (CYP) genes → increases mRNA expression → increases the synthesis of CYP proteins
- A greater number of CYPs are available to metabolize the drug substrate → increases the *rate and extent* of metabolism of the substrate.
- Increased rate of drug metabolism means a greater percentage of the dose of the drug is metabolized → less drug reaches the systemic circulation → ↓ drug concentration at the site of action → the therapeutic effect may be reduced

- ❖ **Remember:** Strong CYP3A4 inducers are frequently strong inducers of P-glycoprotein. The two proteins are frequently co-expressed in tissues like the small intestine and the liver and have overlapping substrate specificity. This means they share many of the same substrates and work together in eliminating xenobiotics like drugs and toxins.
- Induction of P-glycoprotein increases the concentration of the efflux pumps, which increases the rate and extent of efflux of the drug substrates.**

Physiology: Tissues that express drug metabolizing enzymes

The liver is the primary site of metabolism. Genes for the various metabolic enzymes are highly expressed.

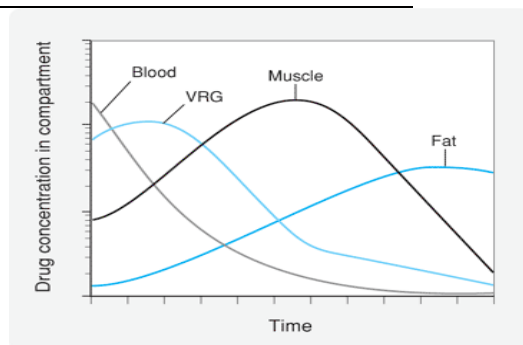
CYP3A4 is also highly expressed in intestinal epithelial cells, which contributes to first-pass metabolism of oral drugs.

Kidney, lung, skin, brain, and placenta also express genes for metabolic enzymes to some extent.

12. A trauma patient in the ED has a seizure. An injection of a highly lipophilic antiseizure drug with an almost immediate onset of action is administered. The duration of antiseizure effect is less about 20 minutes even though the half-life of the drug is over 24 hours.

What is the pharmacokinetic principle that best explains the short duration of action despite the long half-life of the drug?

Hint: The chart on the right



Don't forget to explain the mechanism out loud to yourself, someone else in the room, or your pet.

Rationale: Redistribution

Redistribution is the phenomenon in which the rapid initial distribution of a dose of an intravenous highly lipid soluble drug to the brain. Over time, the drug redistributes to less vascularized tissues that have greater tissue mass, such as muscle and fat, leading to a decline in the concentration in the CNS.

Thinking it through:

- The drug must be highly lipophilic in order to rapidly cross the blood-brain barrier.
- Drug molecules rapidly distribute through the blood from the injection site in the vein to organs with high blood flow (brain, heart, kidneys, liver).
- Drug molecules moves down their concentration gradient from the blood (area of high concentration initially) to brain tissue.
- As the drug redistributes to and is taken up by less vascularized tissues that have greater tissue mass, such as muscle and fat, blood levels in the brain vasculature decrease → the drug moves down its concentration gradient from the area of high concentration in brain tissue back into the blood.
- The effect of the drug is terminated.

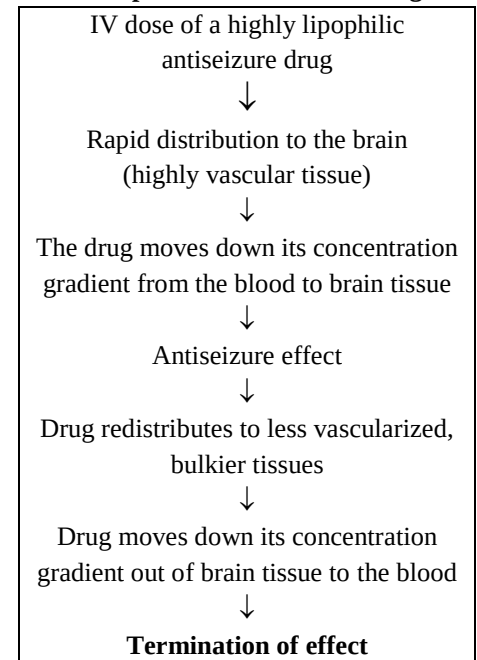
The drug will be metabolized and excreted according to its physicochemical properties.

- ❖ As the plasma concentration decreases, drug moves out of the peripheral tissues back into the circulation. It distributes to the CNS again in much lower concentrations.

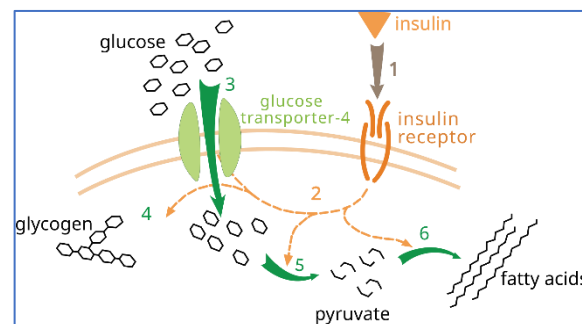


Practice point: Even after the drug is withdrawn, the patient could still feel drug effects, such as drowsiness, dizziness, lack of concentration.
Gradually over time the drug leaves the body.

Application of redistribution to the example of an antiseizure drug



13. A 45-year-old woman injects insulin subcutaneously to manage her diabetes mellitus. Insulin is a protein that distributes in blood, diffuses into tissues, and activates its receptor tyrosine kinase on plasma membranes. Second messenger signaling causes the GLUT4 transporters stored in intracellular vesicles to be inserted in the plasma membrane of hepatocytes, striated (skeletal and cardiac) muscle, and adipose tissue. Glucose passively enters the cell.



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- What type of transport mechanism is described?
 - The glucose transporter is classified in what superfamily of transport proteins?
- Short answers.

Rationale: Facilitated diffusion / SLC transporter superfamily

Glucose is taken up into cells by facilitated transport down its concentration gradient. GLUT carrier proteins are of the solute carrier superfamily, specifically SLC2 family.

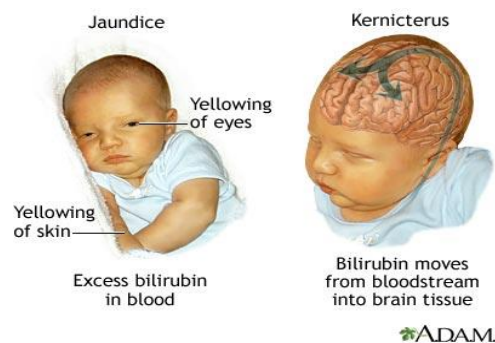
FYI: Several types of GLUT facilitated transporters and two types of SGLTs secondary active transporters (with Na^+) have been identified.

Brainteaser

14. A baby born 3 weeks early requires antibiotic therapy. Sulfonamide antibiotics should be avoided in because their use can lead to accumulation of unconjugated bilirubin in the brain, which can cause encephalopathy and irreversible neurological damage.

What is the mechanism of the drug-bilirubin interaction leading to this adverse effect?

- Blocking bilirubin hepatic excretion
- Displacing bilirubin from albumin binding sites*
- Inhibiting the efflux pump at the blood-brain barrier
- Inhibiting bilirubin conjugation to form the water-soluble metabolite



❖ Information to help you identify clues and think through the question.

Bilirubin

- Bilirubin is a highly lipophilic breakdown product of hemoglobin in aging red blood cells.

- “Unconjugated” bilirubin binds albumin in the blood for transport to the liver.
Remember: Substances bound to plasma proteins do not cross physiologic barriers.
- In the liver, bilirubin is conjugated with glucuronic acid (phase 2 reaction) forming water soluble “conjugated bilirubin”, which is excreted mainly through the bile.

Newborns have immature systems:

- Newborns, especially premature babies, have immature metabolic systems, including low-functioning UDP-glucuronyltransferase (UGT), renal excretory system, and a relatively more permeable blood-brain barrier.

Sulfonamide antibiotics PK properties:

Notice the PK property related to the information on bilirubin disposition.

A: oral / I.V.

D: Highly bound to plasma albumin

M: N-acetylation

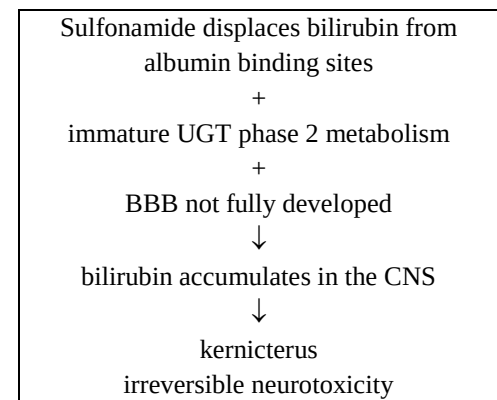
Bioavailability: ~100%

E: urine; metabolites and unchanged drugs

Rationale: B. displacement from albumin binding sites

When unconjugated bilirubin exceeds the binding capacity of albumin and the metabolic capacity of the liver → neurotoxicity

- Sulfonamides have a high affinity than bilirubin for albumin in plasma. Thus, sulfonamides can displace bilirubin at albumin binding sites.
- Concentrations of free unconjugated bilirubin in plasma rise.
- Because neonates, especially premature neonates, have immature, more permeable blood-brain barriers, the free unconjugated (lipophilic) bilirubin can pass into the CNS and deposit in brain tissue.
- Severe hyperbilirubinemia can lead to bilirubin-induced neurologic dysfunction – bilirubin encephalopathy known as **kernicterus**.



Incorrect options:

- A. Blocking bilirubin hepatic excretion: Sulfonamides are metabolized by acetylation, a different metabolic pathway and do not interfere bilirubin transport to the bile.
- C. Inhibiting the efflux pump at the blood-brain barrier: Transporters are not mentioned.
- D. Inhibiting bilirubin conjugation to form the water-soluble metabolite: Sulfonamides are metabolized by acetylation, a different metabolic pathway.

15. A 53-year-old woman presents to her family physician with complaints of hot flashes that are so bothersome that they are affecting her work as an elementary school teacher and her sleep. Relevant medical history shows hysterectomy seven years ago for uterine fibroids, no current prescription medication, ibuprofen as needed for headache or musculoskeletal pain, and no known drug allergies. The patient agrees to estrogen therapy for the menopausal symptoms and for bone health. Estrogen has a high first-pass effect. The patient opts for the transdermal form of drug delivery.

i) What are some advantages of the selected route of delivery?

- A. Gastric side effects are avoided.
- B. Lower doses are effective.
- C. Reduced dosing frequency improves patient adherence.
- D. Sustained drug levels reduce peaks and troughs.
- E. All of the options*

ii) What property is essential for transdermal drug delivery?

Rationale: E. all options / lipophilicity

Advantages of transdermal drug delivery:

- 1- Convenient, painless drug delivery
- 2- The drugs are not subject to harsh gastric environment or first-pass metabolism.
- 3- Maintenance of therapeutic effect overnight because of gradual delivery of drug from the patch
- 4- Decreased incidence and/or intensity of both undesired effects

Examples of drugs administered transdermally:

Nicotine for smoking/tobacco use withdrawal; scopolamine for motion sickness; nitroglycerin for angina pectoris; testosterone and estrogen for hormone replacement therapy; various estrogens and progestins for contraception; fentanyl for pain relief; and others

Disadvantages:

- 1) Slow onset of action owing to gradual release of drug from the drug reservoir
 - 2) The patch may be irritating to the skin.
-

16. An HIV patient's antiretroviral therapy includes drugs called oral nucleoside analogs (they are structurally similar to natural nucleosides, the building blocks of DNA and RNA). The drugs are inactive until they are activated by triphosphorylation by host kinases within the infected white blood cells where they act to prevent key viral processes.

Based on this description, the parent compound (the drug in the pills the patient swallows) is called a/an:

- A. Antagonist
- B. Inverse agonist
- C. Prodrug*
- D. Xenobiotic*

Rationale: Prodrug and xenobiotic are both correct answers.

A prodrug is a compound that is inactive on its own. It must be converted (biotransformed, metabolized) to the active compound. This process occurs in the liver with most prodrugs. However, conversion to the active form can happen in the blood or tissue cells, as in this case. In the example given, the drug is activated by phosphorylation inside the white blood cells using the cells' own kinases.

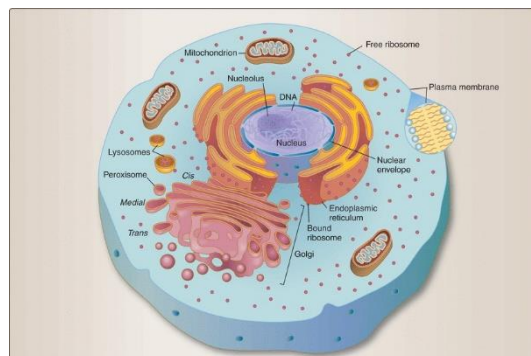
A xenobiotic is a substance foreign to the body. The word is derived from the Greek: *xenos* for foreign or stranger and *bios* meaning life. Almost all drugs are xenobiotics.

Incorrect options: Antagonist / Inverse agonist are incorrect. It is the active metabolite that produces the effect (competitive antagonist). The parent drug has no therapeutic activity.

17. An adolescent female is treated for whooping cough (pertussis) with the antibiotic clarithromycin. The drug undergoes hepatic biotransformation via the cytochrome P450 monooxygenase system (CYP3A4 to be precise).

In what region of the hepatocyte does this reaction occur?

- A. Endoplasmic reticulum*
- B. Golgi apparatus
- C. Lysozymes
- D. Mitochondria
- E. Plasma membrane
- F. Ribosomes



Rationale: Smooth endoplasmic reticulum

The CYP450 drug-metabolizing enzymes and UDP-glucuronyltransferase (UGT) are located in the lipophilic endoplasmic reticulum membranes of the liver and other tissues.

18. A 22-year-old male with schizophrenia resistant to first-line antipsychotic agents is admitted to the psychiatric hospital and initiated on treatment with clozapine, which has been shown to be effective for this indication. The drug is effective but considered a last-line agent due to its toxicities apparently caused by a toxic metabolite: life-threatening agranulocytosis, myocarditis, seizures. The patient smokes two packs of cigarettes per day (10 pack years) but will not be able to smoke while in the psychiatric hospital. Because the patient is a smoker, the initial dosage of the drug will be adjusted upward according to evidence-based recommendations and readjusted as needed. The patient will be closely monitored for signs and symptoms of toxicity.

Which of the following is the primary metabolic enzyme family involved in this drug interaction?

- A. CYP1A family*
- B. CYP2B family
- C. CYP2C family
- D. CYP2D family
- E. CYP3A family

Definitions:

Agranulocytosis: Marked decrease in granulocytes, a type of white blood cells of the innate immune system

Myocarditis: Inflammation of the heart muscle

Seizures: Sudden uncontrolled electrical disturbance in the brain.

Rationale: A. CYP1A family

The CYP1A family of cytochrome P450 enzymes, CYP1A1 and 1A2, metabolize some polycyclic aromatic hydrocarbons to carcinogenic intermediates.

Clozapine is a substrate primarily of CYP1A2. Smoking induces CYP1A2.

Cigarette smokers metabolize clozapine more rapidly than nonsmokers. Smokers may require twice the daily dose as nonsmokers to achieve equivalent plasma concentrations of clozapine. As time goes by, the concentration of drug metabolizing enzymes tends to slowly decrease, which may require a lower dose of the clozapine.

If the patient starts smoking again after being discharge, doses will then need to be readjusted.

19. A 70-year-old person (pronouns he/him/his) with recurrent metastatic colorectal cancer is evaluated for treatment with a particular chemotherapy drug that is converted to an active metabolite. This metabolite is inactivated by an enzyme that catalyzes the transfer of glucuronic acid from cofactor UDP-glucuronic acid to the active molecule. The patient's medical history reveals that he has a genetic disorder (Gilbert's syndrome), in which a gene mutation causes a deficient activity of the metabolic enzyme necessary for catalyzing the inactivation of this drug. What class of enzymes inactivates this drug? (Short answer)

Rationale: Phase 2: Transferase

Glucuronidation is a phase 2 reaction. UGT is a transferase that catalyzes conjugation reactions.

➤ **Learning point:** This case is an example of pharmacogenomics – genetic polymorphisms affecting an individual's response to drugs.

The scenario describes the phase 2 reaction catalyzed by UDP-glucuronosyl transferase (UGT)1A1. The potential result of the deficient UGT1A1 activity is higher concentrations of

drug for a given dose, resulting in chemotherapy toxicity – in the case of the drug described in this vignette, severe diarrhea requiring hospitalization and bone marrow suppression.

20. A 57-year-old woman with a long history of hypertension presents to her cardiologist with complaints of fever, muscle and joint pain, and a rash on her face. Included in her medication regimen is a vasodilator metabolized by N-acetyltransferase type 2. In physical examination, a maculopapular malar rash (butterfly rash), generalized lymphadenopathy, and splenomegaly are noted. Results of the work-up reveal drug-induced lupus-like syndrome. The offending antihypertensive drug is discontinued. The symptoms and signs gradually resolve. Which one of the following describes the biotransformation reaction that produced the toxic effect?

- A. Conjugation*
- B. Hydrolysis
- C. Oxidation
- D. Reduction

Hint: Notice the mode of metabolism of the offending drug.

Rationale: Conjugation: Acetylation of the N-terminal by N-acetyltransferase (NAT2) is a phase 2 conjugation reaction.

The vignette describes conjugation of hydralazine with an acetyl group obtained from acetyl CoA. Individuals with pharmacogenetic deficiency in NAT2 function are at increased risk for developing the immunologic lupus-like reaction.

21. A 52-year-old woman takes conjugated estrogens for the management of menopausal vasomotor symptoms (hot flashes). Estrogen metabolism is a complex process that includes the formation of sulfate and glucuronide conjugates, which are excreted via the biliary system. When she eats a meal, the gallbladder releases its contents, including the conjugated metabolites, into the duodenum. The metabolites travel through the small intestine to the ileum, where they are hydrolyzed by bacteria in the gut flora. The reconstituted estrogen is then passively reabsorbed, which contributes to the plasma concentration of the hormone.

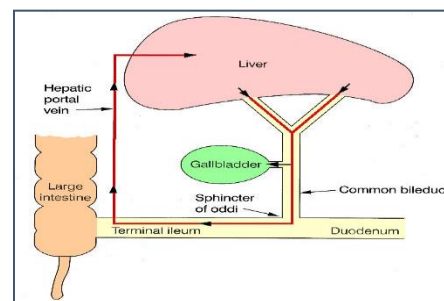
What pharmacokinetic process is described?

Short answer.

Rationale: Enterohepatic circulation.

Enterohepatic circulation can also increase the systemic levels and half-life of several drugs.

- **Physiology of bile acids:** Bile acids are necessary for the digestion and absorption of fats and fat-soluble vitamins in the



small intestine. Bile acids are synthesized in the liver from cholesterol, conjugated, and secreted into the gallbladder. Fatty foods stimulate gallbladder contraction, which release its contents into the common bile duct, which drains into the duodenum. At the terminal ileum, hydrolysis of conjugated bile acids (bile salts) and reabsorption of bile acids occurs.

22. A critically ill patient receiving I.V. gentamicin (antibiotic) for a serious gram-negative bacterial infection complains of ringing in her ears and dizziness; nystagmus is noted on physical exam. The signs and symptoms are characteristic of gentamicin toxicity. Her gentamicin blood levels are found to be in the toxic range.

Gentamicin properties:

- Gentamicin is a small molecule, highly polar drug.
- It is not absorbed from the GI tract, therefore given only parenterally.
- Only a small percentage of the dose binds to plasma proteins.
- It is not metabolized.
- It is a small molecule that passes through the glomerulus.

Definition of nystagmus: Repetitive uncontrolled eye movements.

i) Impairment of what organ could lead to the toxic serum levels?

Hint: The answer can be discerned from the drug's pharmacokinetic properties.

ii) What product in serum is a useful biomarker of the function of this body system?

Hint: Page 27 on the Notes Handout

Short answers.

Rationale: Impaired renal function would reduce the clearance of the drug through the urine.

Elevations in serum creatinine indicates that glomerular filtration rate, therefore kidney function, is reduced.

1. The drug is a small molecule solute in plasma and it is mainly unbound. It passes through the fenestrations of the glomerular capillaries in the plasma – that is, it is filtered by the glomerulus; it is not secreted.
2. Creatinine in the serum is largely (but not completely) filtered by the glomerulus and it is not reabsorbed in the nephron. Accumulation of creatinine in the serum indicates that glomerular filtration is decreased, which is why it is a useful biomarker of renal function.
3. When glomerular filtration rate is decreased, renal function is decreased. Gentamicin can accumulate in plasma, leading to gentamicin toxicity.

Adverse effects are common with gentamicin therapy.

23. A 65-year-old man is treated for non-Hodgkin lymphoma with a chemotherapy regimen that includes methotrexate, which has potentially serious toxicities and a narrow therapeutic window. The patient is counseled on how to recognize the adverse effects and the drug interactions to avoid. Included is avoidance of NSAIDs use, which can inhibit the secretion of methotrexate into the proximal tubule, which would increase the risk of methotrexate toxicity.

Methotrexate PK properties:

A: IV for the treatment of cancer

D: moderate plasma protein binding (~50%)

M: not metabolized by CYPs or conjugation

E: renal glomerular filtration and secretion; ~90% of the dose is excreted in urine as unchanged drug

Definition, NSAID: nonsteroidal anti-inflammatory drug, such as ibuprofen (Advil, Motrin) and naproxen (Aleve)

What is the likely pharmacokinetic mechanism of the methotrexate-NSAIDs interaction?

- A. Allosteric potentiation
- B. Competitive inhibition of OAT*
- C. Inhibition of metabolism
- D. Ion trapping

Hint: Think about what can happen when the portals are blocked.

Rationale: Inhibition of transport into the proximal nephron.

The student can deduce that inhibiting the secretion of methotrexate in the proximal tubule increases the risk of toxicity by inhibiting excretion of active (unchanged) drug, leading to elevated serum drug levels and accumulation of drug in tissues (toxicity).

24. Excretion of unchanged nicotine is pH-dependent. At a urinary pH of 7 or greater, excretion of unchanged nicotine is approximately 2%, whereas an average of 23% is recovered unchanged at a urinary pH of less than 5.

Chemically, what type of drug is nicotine?

Short answer

Rationale: Nicotine is a weak base.

In acidic pH, the conjugate acid BH^+ is favored. This polar ion is trapped in the urine – not reabsorbed, therefore “recovered in the urine”.

At the higher pH, a greater proportion of the nicotine is in the nonionized state, ie, the lipophilic base, B form, which is passively reabsorbed into the systemic circulation (not recovered = not excreted in the urine).

25. A 38-year-old woman complains to her doctor that she has been feeling sad, hopeless, and irritable for about 6 months. She has lost interest in everything. She relates that her family and co-workers have been urging her to get help. After a full workup, she is given a diagnosis of major depressive disorder. An appointment is made with a mental health counselor and an electronic prescription (e-prescription) is submitted for an antidepressant that inhibits the reuptake of serotonin. The drug is taken orally once daily. It is 95% plasma protein bound. If such a large percentage of the drug is protein bound, how does it achieve and maintain adequate concentrations at the site of action in the brain?

Hint: Notes page 16.

Rationale:

Lipophilic drugs are typically highly bound to plasma albumin and α 1-acid glycoprotein.

Although a drug may be highly bound to plasma proteins, the small fraction of unbound drug can enter the CNS and produce its effects.

At steady state, there is a **dynamic equilibrium** between bound and unbound drug.

As free drug enters the CNS, the concentration of free drug in plasma is reduced.

More drug dissociates from the plasma proteins to **maintain the dynamic equilibrium**, allowing sustained amounts of free drug to move to the site of action.

- ❖ A dynamic equilibrium also exists between drug bound to tissue proteins, lipids, and cellular components and unbound drug in interstitial fluids. As free drug moves between tissues and plasma, bound drug dissociates to maintain dynamic equilibrium.

Core principle of pharmacokinetics and drug

=====

SUMMARY

Pharmacokinetics: The branch of pharmacology that deals with the effects of biologic systems on drugs (“what the body does to the drugs”). The process involved in pharmacokinetics are absorption, distribution, and elimination (metabolism=biotransformation and excretion) (ADME processes).

Drugs must be able to

- Withstand digestive fluids and enzymes (drugs administered to the GI tract)
- Cross physiological barriers
- Survive metabolic process → active drug in blood stream
- Distribute to the site of action in therapeutic concentrations
- Exit the site of action after providing its specific effect
- Undergo elimination

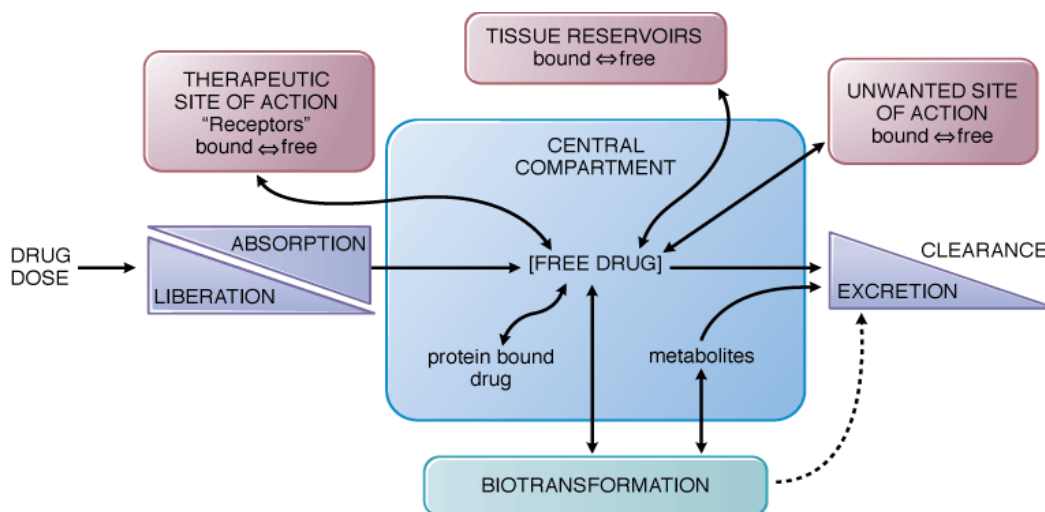
metabolism to more polar compounds, and/or

excretion of unchanged drug and/or metabolites

- The physicochemical characteristics of a drug determine its ability to withstand digestive fluids and enzymes and be absorbed from the GI tract, cross physiologic barriers, survive

metabolic processes, distribute to the site of action in therapeutic concentrations, exit the site of action, and be eliminated.

- Drugs move along a concentration gradient by passive diffusion across cell membranes or carrier-mediated transport. Drugs may also be extruded by energy-dependent active transport. Very impermeant drugs may enter / exit cells by endocytosis / exocytosis.
- The rate and extent of flow of drug molecules are influenced by the magnitude of the concentration gradient across the membrane, lipid-water partition coefficient, and membrane surface area, thickness, and permeability.
- Most drugs are weak acids or bases that are present in solution as both the non-ionized and ionized species. The non-ionized molecules usually are more lipid soluble and can diffuse readily across the cell membrane, whereas ionized molecules usually are less able to penetrate the lipid membrane because of their low lipid solubility. Transmembrane distribution is determined by the drug's pK_a and the pH gradient across the membrane.
- Different routes of administration are used in clinical medicine various reasons, such as convenience, rapid onset of action, maximizing concentration at the site of action, avoiding the first-pass effect, and minimizing adverse effects.
- The rate and extent of drug distribution are determined by physicochemical properties of the drug, and the physiologic factors of cardiac output, regional blood flow, capillary permeability, and tissue volume. Extensive binding to inert proteins and other macromolecules may slow the rate at which a drug reaches the site of action and may prolong the duration of action.



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*: www.accessmedicine.com
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- Biotransformation is a complex process, involving numerous enzyme systems, that usually detoxifies xenobiotics; however, metabolic activation may occur.

- Phase 1 reactions (oxidation, reduction, or hydrolytic reactions) add or expose functional groups of lipophilic xenobiotics, which can inactivate them or alter their biologic properties.
- The phase 1 cytochrome P450 (CYP) enzymes biotransform most xenobiotics; CYP3A4 is the most abundant isoform, metabolizing almost 50% of all drugs.
- Phase 2 reactions catalyze the transfer (conjugation) of a functional group from a cofactor onto the parent compound or phase 1 metabolite, which enhances hydrophilicity and renal elimination.
- CYP enzyme activity is quite variable due to genetic polymorphisms, genetic changes that exist in at least 1% of the human population.
- Enzyme activity also varies due to concurrent administration of other drugs, which result in **induction and inhibition** of the drug metabolizing enzymes.
- The most common method of induction is a nuclear receptor-mediated increase in gene transcription.
- Inhibition of CYP enzymes is the most common cause of harmful drug-drug interactions. The most common mechanism is competitive inhibition.
- Drug disposition is also affected by membrane transport proteins, which mediate the cellular uptake and efflux of xenobiotics.
- Oral drugs may undergo first pass metabolism as they are exposed to drug metabolizing enzymes in the GI tract and the liver prior to reaching the systemic circulation.
- Some drugs undergo enterohepatic circulation in which substances secreted in the biliary system are extruded into the GI tract during a meal, and are reabsorbed by enterocytes and transported back to the liver.
- Most drugs are excreted in urine, either as the parent compound (unchanged drug) or as metabolites with polar characteristics.
- The processes involved in renal elimination are glomerular filtration, active tubular secretion, and passive tubular reabsorption.
- Only unbound drugs are filtered by the glomerulus; lipophilic drugs undergo passive reabsorption.
- Inter-individual variations can affect the rate and extent of drug elimination.
- Estimated GFR is used to assess kidney function. Creatinine is a marker of kidney function.

Resources:

Name of Discipline/Organ System: **General Pharmacology** > Principles of Pharmacology

Name of Brick: Introduction to Pharmacology > Foundations and Frameworks

Link: <https://exchange.scholarrx.com/brick/pharmacology-foundations-and-frameworks> Scholar

Rx Bricks:

Pharmacokinetics and Pharmacodynamics > Pharmacokinetics: Drug Transport and Diffusion

Link: <https://exchange.scholarrx.com/brick/drug-transport-and-diffusion>

Name of Brick: Pharmacokinetics and Pharmacodynamics > Pharmacokinetics: Drug Administration, Metabolism, and Excretion

Link: <https://exchange.scholarrx.com/brick/drug-administration-metabolism-and-excretion>

Cell and Molecular Biology > Cellular Biology

Name of Brick: Transport Across Membranes <https://exchange.scholarrx.com/brick/transport-across-membranes>

Name of Brick: Cell Signaling <https://exchange.scholarrx.com/brick/cell-signaling>